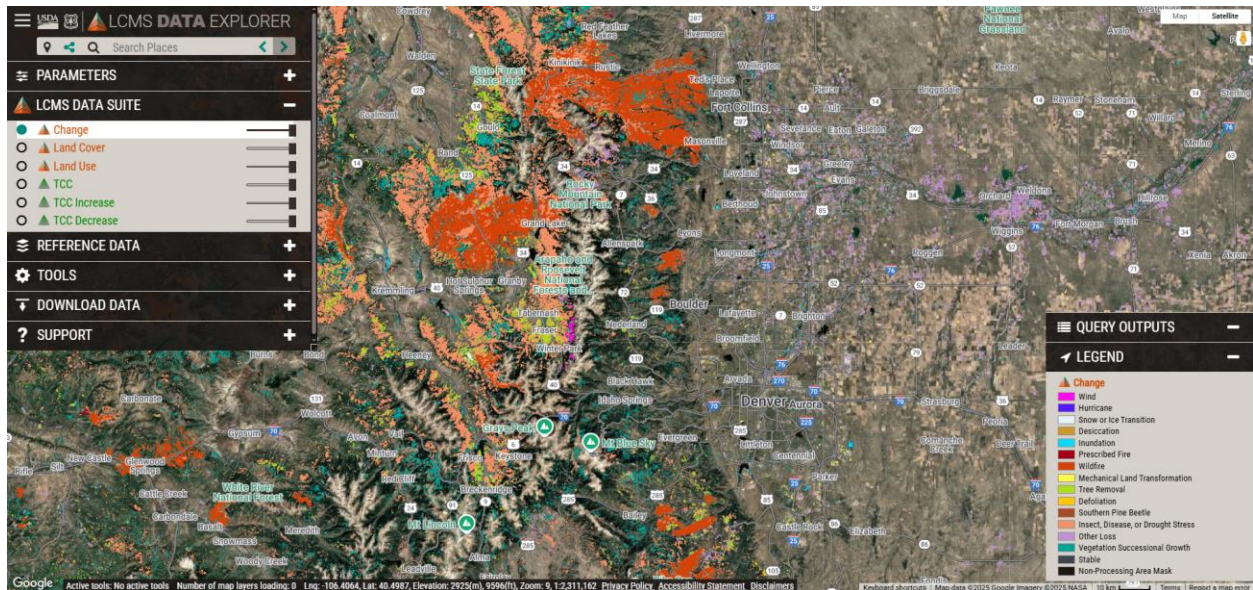


OVERVIEW: LCMS Data Explorer



Introduction

This guide provides an overview for navigating and downloading data produced for the USDA Forest Service's Landscape Change Monitoring System (LCMS) using the LCMS Data Explorer. The LCMS Data Explorer is the best tool for quickly viewing and downloading available data products. Tree Canopy Cover (TCC) data can also be viewed and downloaded in the LCMS Data Explorer.

The LCMS Data Explorer can be found at: <https://apps.fs.usda.gov/lcms-viewer/>

Objectives

- Learn how to navigate the LCMS Data Explorer web application.
- Download data from the LCMS Data Explorer.

Required Software

- A web browser other than Microsoft Internet Explorer (Tested on Chrome, Firefox, and Edge).

Prerequisites

- Basic understanding of how to use a simple GIS, such as Google Maps or Google Earth.

Contacts

- LCMS general inquiries: sm.fs.lcms@usda.gov

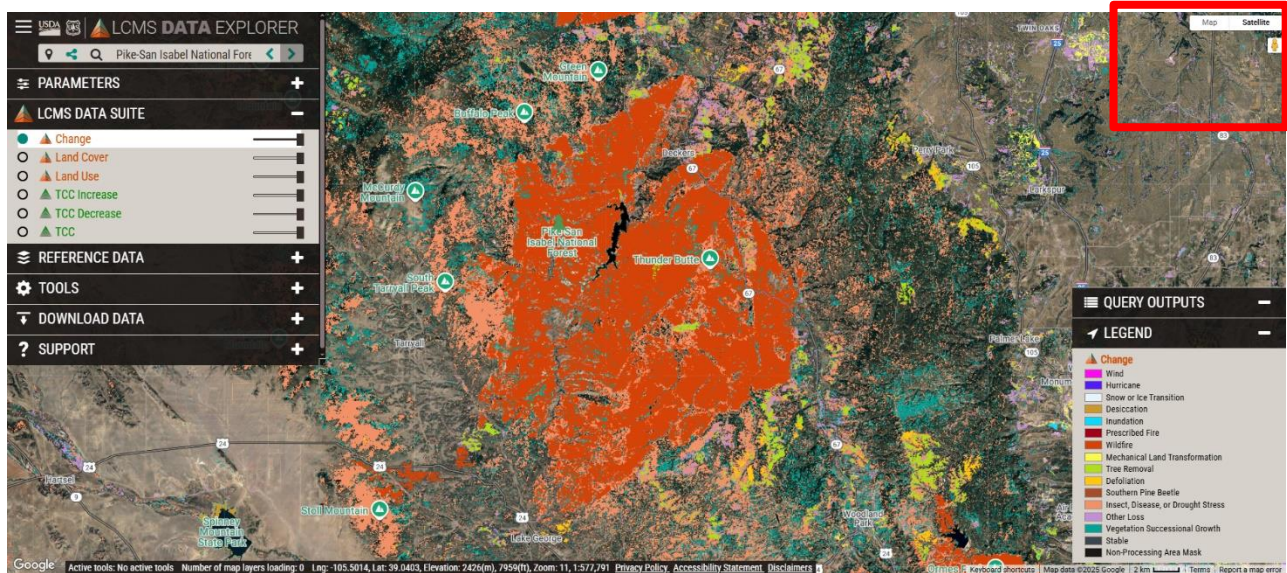
Table of Contents

Part 1: Navigating the LCMS Data Explorer	3
A. Explore the Google Tools	3
B. Viewer Information Ribbon & Legends.....	3
Part 2: Menu Options & Tools.....	5
A. Title and Search Bar	5
B. Parameters tab	5
C. LCMS Data Suite tab.....	7
D. Explore the Reference Data tab	10
E. Explore the Tools tab	10
Part 3: Download Data & Support Tabs.....	17
A. Download Data tab	17
B. Support tab	17

Part 1: Navigating the LCMS Data Explorer

This section introduces the user to general navigation and functionality of the LCMS Data Explorer.

A. Explore the Google Tools



Launch the LCMS Data Explorer by [clicking this link](#). The tools along the upper-right side of the browser window are the default tools within all web mapping applications using Google Maps.

1. At the top right you can choose between **map** view and **satellite** view for your base layer.
 - i. **Map** view has a checkbox drop-down option to turn terrain view on and off.
 - ii. **Satellite** view has a checkbox drop-down option to turn labels on and off.
2. Below the full screen button are zoom tools for viewing the map at different scales.
 - i. Zoom In and Out can also be accomplished using the scroll wheel on your mouse.
3. The orange person icon allows you to drop into street view.



- i. You can click and drag the person onto the map area. Any area that becomes highlighted in blue is available for you to drop the person to see surface level imagery in that area.
- ii. To exit street view, click the left-pointing arrow in the upper left corner of the window.

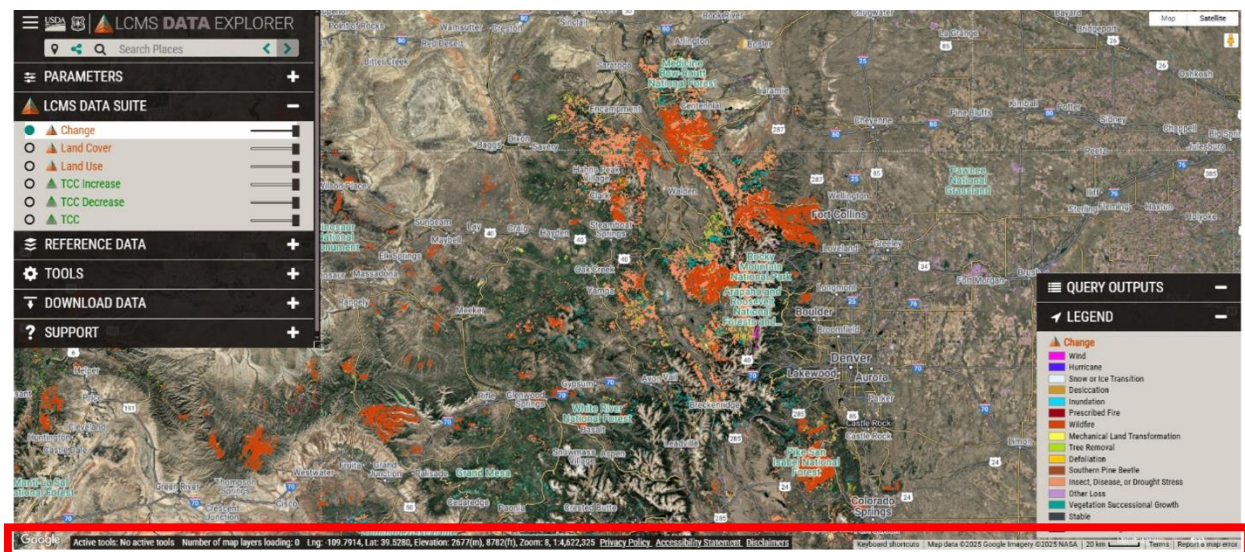
B. Viewer Information Ribbon & Legends

1. For additional information on a tool or layer, hover the mouse over the name and additional text will appear (if available).
2. Navigate to the ribbon of text along the bottom of the data viewer.

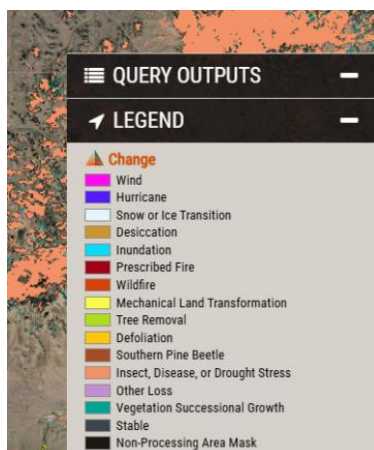
The info bar contains various bits of information pertaining to the viewer and the data

- **Google:** A link to the current view in Google Maps

- **Currently active tools:** Which tools are turned on.
- **Queue length for maps from GEE:** How many data layers still need to load.
- **Number of map layers loading tiles:** How many of the enabled layers that are actively loading tiles.
- **Lat/Long, Elevation:** Based on the mouse location.
- **Zoom, Scale:** Map view zoom level and effective scale.
- **Privacy Policy, Accessibility Statement, and Disclaimers:** U.S. Department of Agriculture, Forest Service information.
- **Google Maps Information:** Google Maps source, scale, and terms



3. The query outputs and legend, located on the bottom right of the viewer, will display query results and information about LCMS and Reference Data products that are turned on.

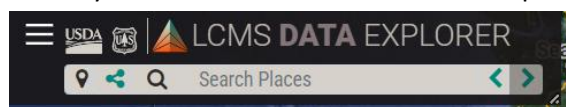


Part 2: Menu Options & Tools

This section reviews the menus and options available and analysis tools for exploring the LCMS data.

A. Title and Search Bar

1. The top of the left side of the browser window is always visible. It allows the user to:
 - i. Minimize the rest of the left sidebar
 - ii. Go to the LCMS Home Page by clicking on the U.S. Department of Agriculture, Forest Service logo
 - iii. Center the map on your current location
 - iv. Share your current instance of the Data Explorer by clicking the teal tri-lobed icon.



- v. Search for a place
- vi. Go to prior map views

B. Parameters tab

The menus and associated options are located along the left side of the browser window. To show the options under each heading, maximize the menu tabs by clicking the “+” on the right side of the tab. When you are ready to minimize the tab, simply click the “-”.

1. **Add LCMS Time Lapses:** Choose “Yes” if you would like to view an interactive time lapse of LCMS change, land cover, land use products. These time lapses are created on-the-fly and can be very slow to load, but provide a very useful method for visualizing LCMS products across time. If “Yes” is selected, only Standard mode is valid in mode selection (see below).

PARAMETERS

Add LCMS Time Lapses:

Choose which mode:

Change: Stable color

Choose analysis year range:

1985 - 2024

LCMS Classification Level Selection

Change Classification Level

☐ 1
 ☐ 2
 ☒ 3

Land Cover Classification Level

☐ 1
 ☐ 2
 ☒ 3
 ☐ 4

Land Use Classification Level

☐ 1
 ☒ 2

2. Choose which mode:

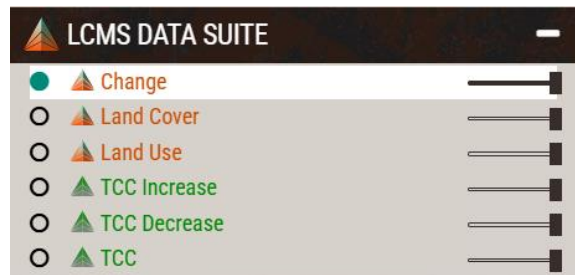
- i. **Standard** mode will load all core LCMS layers.
 - ii. **Advanced** mode will load a five-year range summary starting with the start year and a five-year summary ending with the end year of the selected analysis year range. The year range selected must be at least six years to detect a difference between the start and end years. Advanced mode is explained in greater detail below in the LCMS Data Suite tab, ii. Advanced mode section.
3. The analysis year range can be modified from “1985-2024”. This year range applies to the time lapses or the “Advanced mode” analysis range.
 4. LCMS uses a hierarchical (nested) legend to represent different levels of detail for its Land Cover, Land Use and Change products. Choose the appropriate balance of accuracy (lower levels typically exhibit higher accuracies) and level of thematic detail for your needs using the options available under the “LCMS Classification Level Selection”.
 5. Once you have finished changing any parameters, refresh the maps by clicking the **Submit** button.
 - i. If the Submit button is disabled, either the selected options are already being processed, or other layers are loading, and you will need to wait for them to finish.

C. LCMS Data Suite tab

1. To explore the LCMS products, click on the **LCMS Data Suite** tab located under the Parameters tab.
2. Hover over each layer name for additional text that describes the layer.
3. The following products will be shown as options to toggle on/off depending on what parameters you choose:

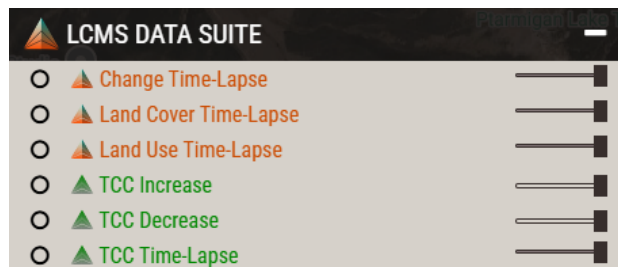
i. Standard mode (default)

- (a) *Change*: Vegetation cover or landscape change from a long-term trend event (e.g drought stress, tree mortality from insects or disease, snow/ice transition) or from an external event (e.g fire, wind/hurricane, tree removal, mechanical land transformation). Each pixel shows its lowest (top-most class in the legend) Change type observed within the selected year range. For example, “Stable” pixels experienced no other changes during the year range, while “Wind” pixels indicate Wind was the lowest Change class number that occurred at that pixel, even if other changes were also present during the selected time range.
- (b) *Land Cover*: most common land cover class for the year range.
- (c) *Land Use*: most common land use class for the year range.
- (d) *TCC Increase*: total of all increases in Tree Canopy Cover (TCC) that falls within LCMS disturbance or vegetation growth areas.
- (e) *TCC Decrease*: total of all decreases in Tree Canopy Cover (TCC) that falls within LCMS disturbance or vegetation growth areas.
- (f) *TCC*: average percentage of Tree Canopy Cover (TCC) over a pixel (1985-2023).



If LCMS Time Lapses were added, additional layers include:

- (a) Change Time-Lapse
- (b) Land Cover Time-Lapse
- (c) Land Use Time-Lapse
- (d) TCC Time-Lapse



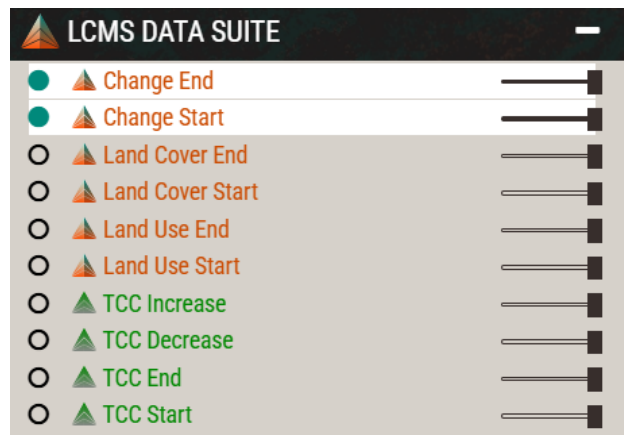
ii. Advanced mode

Additional layers include Start (most common class for first 5 years) and End (most common class for last 5 years) layer of:

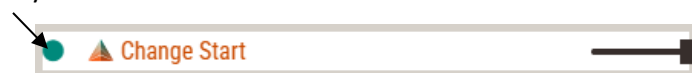
- (a) Change (rather than the mode like Land Cover and Land Use, for each pixel Change will show the class with the lowest class number – I.e. Wind is class number 1, Stable is class number 15 – that occurred during the 5-year start period and the 5-year end period)
- (b) Land Cover
- (c) Land Use
- (d) Average Tree Canopy Cover (from TCC)

As well as the total percent over the year range selected of:

- (e) TCC Increase
- (f) TCC Decrease



4. Turn LCMS product layers on and off by clicking their corresponding radio button to the left of each layer.



5. Use the slider next to each layer to change the transparency in the viewer window.



6. Land cover and land use change can be visualized either by using the time lapses or by toggling between the “start” and “end” land cover and land use layers that are created in advanced mode. Click on the land cover and land use start and end year layers to view the mode (most common) of the five years starting with the start year or ending with the end year, respectively, for land cover or land use.

7. Click on the time lapse layers. When a time lapse layer is activated, all other layers are turned off. They will take a bit to load, so be patient and do not pan around while the maps are loading.



- i. Click the 'Play' button to playback time lapse annual layers.



- ii. Click the "Pause" button to pause the time lapse playback.



- iii. Manually flip through years with the "Back" and "Forward" buttons or drag the year slider (middle slider).

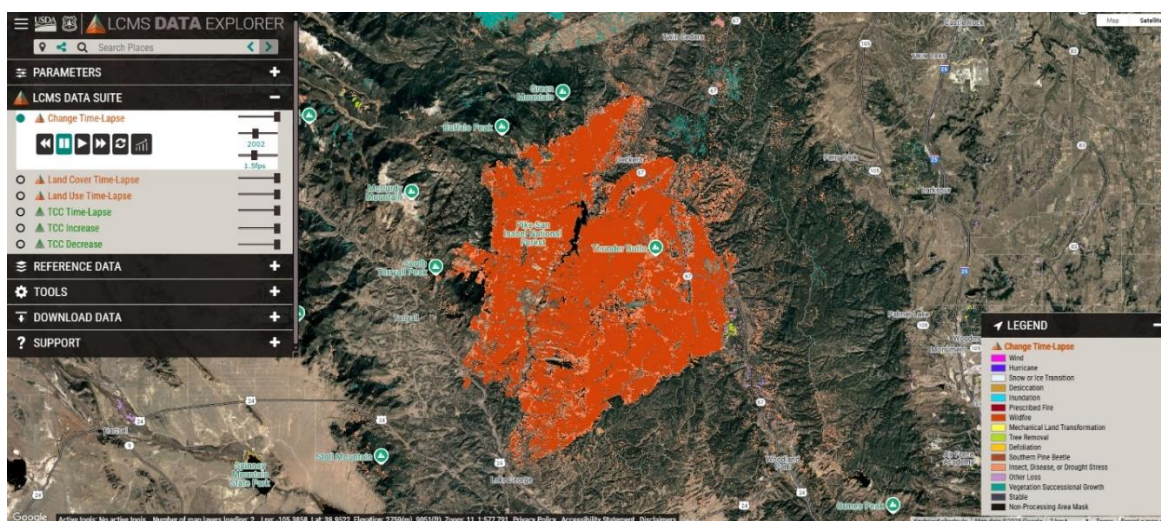


- iv. Adjust the playback speed (1.5fps) with the lower slider.

- v. If tiles fail to load (holes appear in maps) click the "Reload" button.



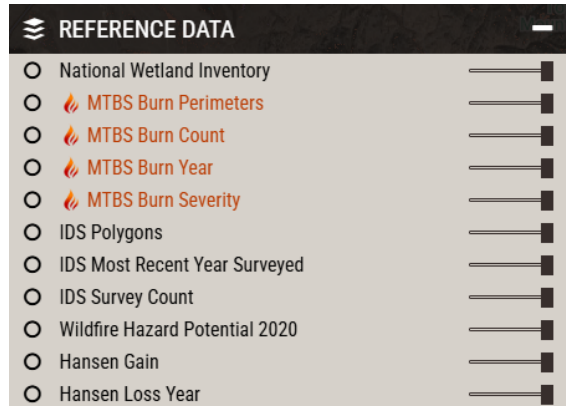
- vi. If you would like to view all layers prior to a given year along with that year, click on the "Ramp Chart" button to toggle cumulative mode.



Example: Hayman Wildfire in Colorado – the large area of contiguous change displayed as Wildfire (deep orange color).

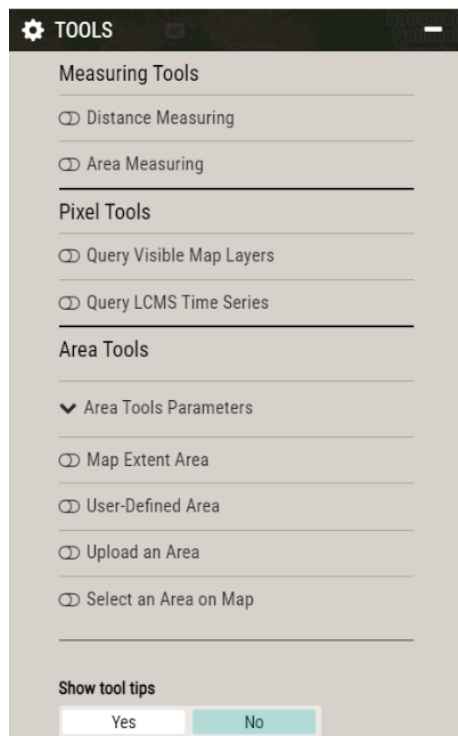
D. Explore the Reference Data tab

1. Next, click the **Reference Data** tab. These reference data sets provide additional geospatial information that may be useful for exploring the LCMS data.
2. This set of reference data allows users to display information directly from the web map without downloading data.
3. Use the sliders to change the transparency of layers on the display window.



E. Explore the Tools tab

1. Click on the plus in the right corner to expand the **Tools** tab.



2. There are three main tools: Measuring, Pixel, and Area. Each tool can be turned on by clicking the slider button located to the left of each tool



3. Measuring Tools:

- i. Distance Measuring: Click anywhere to begin a linear measurement. The option buttons allow the user to undo, delete, and change the color of the measurement line. The user can also choose Imperial or Metric measuring units.

Measuring Tools

☒ Distance Measuring

Click on map to measure distance

Imperial

Metric

↶

🗑️

🖌️

- ii. Area Measuring: Click anywhere to begin delineating an area on the map, **single click** to create a vertex, and **double click** to complete a polygon area. The option buttons allow the user to undo, delete, and change the color of the area boundary line. The user can also choose Imperial or Metric measuring units.

Measuring Tools

☐ Distance Measuring

☒ Area Measuring

Click on map to measure area

Imperial

Metric

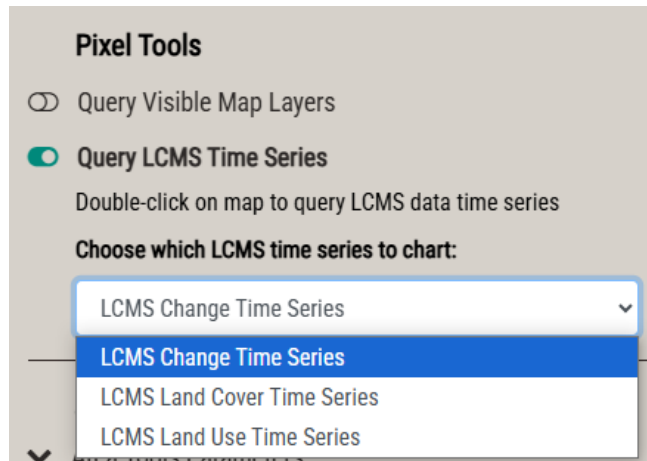
↶

🗑️

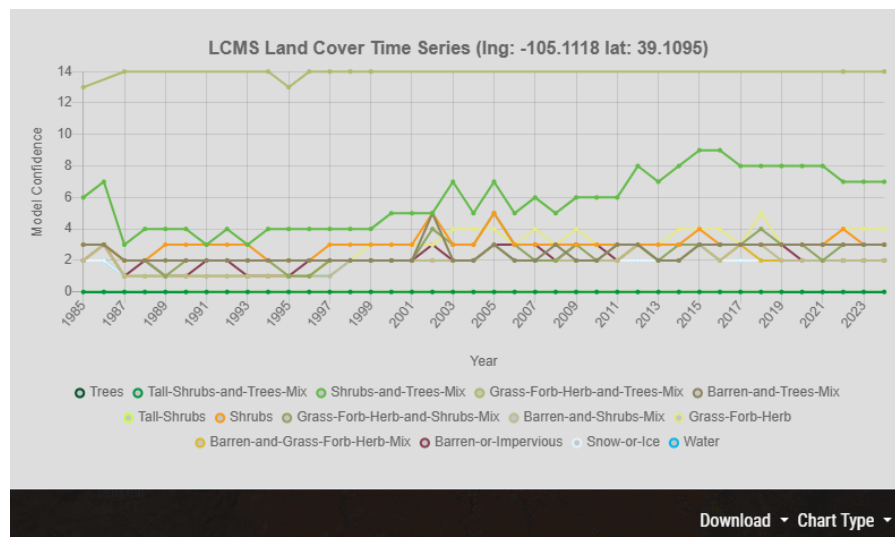
🖌️

4. Pixel Tools:

- i. Query Visible Map Layers:
 - (a) Double-click a point of interest in the project boundary. A table or chart will automatically display the pixel values for visible LCMS or Reference data layers in the “Query Outputs” pane on the right side.
- ii. Query LMCS Time Series layers:



- Choose the LCMS time series to chart by expanding the drop-down menu. Available options are:
 - LCMS Change Time Series
 - LCMS Land Cover Time Series
 - LCMS Land Use Time Series
- Double-click a point of interest in the viewer. A chart will automatically pop-up displaying LCMS product information for that point.
- Turn layers on and off by clicking their legend entries below the chart.
- Note the **Download** button at the bottom right of the chart. Time series charts can be downloaded as a CSV, PNG, or geoJSON.
- Select the **Chart Type** button at the bottom right of the chart to view the time series as either a graph or a table of values.



5. Area Tools: Summarize LCMS products by area (v2024.10 includes a complete rework from v2023.9. The new version is ~10x faster, scales at low zoom levels, and supports multiple charts simultaneously).

- i. **Area Tools Parameters:** The user can select which LCMS products to summarize in the chart by expanding the **Area Tools Parameters** dropdown menu, located just below the **Area Tools** heading.

Area Tools

▲ **Area Tools Parameters**

Area Chart Layers

☐ Tree Canopy Cover

☒ Change Annual

☐ Change Transition

☒ Land Cover Annual

☐ Land Cover Transition

☒ Land Use Annual

☐ Land Use Transition

Area Units

☒ Percentage ☐ Acres ☐ Hectares

1. The LCMS products that are summarized can be selected under the **Area Chart Layers** checkboxes.
2. The units that are used can be changed to portray proportions (“percentage”) of the area, or the actual area as acres or hectares.
3. Any product ending with “-transition” will create a Sankey chart that portrays how classes transitioned. An additional user interface is presented that allows years to be entered to then chart different transition periods. Use the “*plus*” and “*minus*” buttons to add and delete year periods. All periods must be in succession.

Transition Charting Periods

	Period Start Year	Period End Year	
◆	1985	1987	✕
◆	Enter Year	Enter Year	✕
◆	2022	2024	✕
+ -			

- ii. **Map Extent Area:** Will chart any selected product (above) any time the map is panned or zoomed.
- iii. **User-Defined Area:** Allows the user to draw a polygon and obtain LCMS summaries for the area.

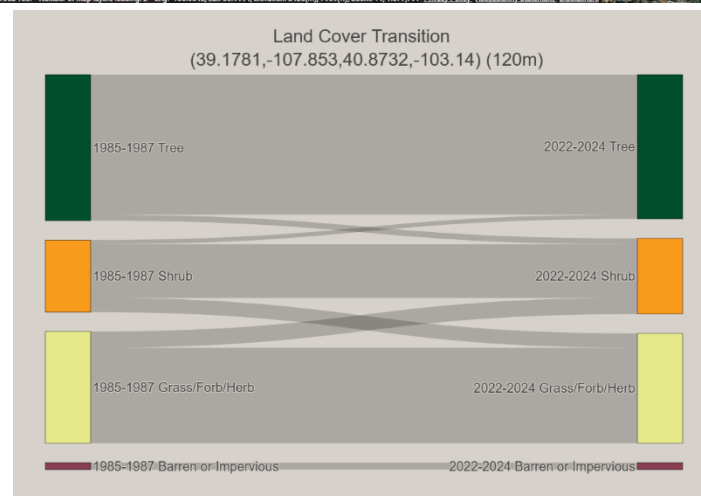
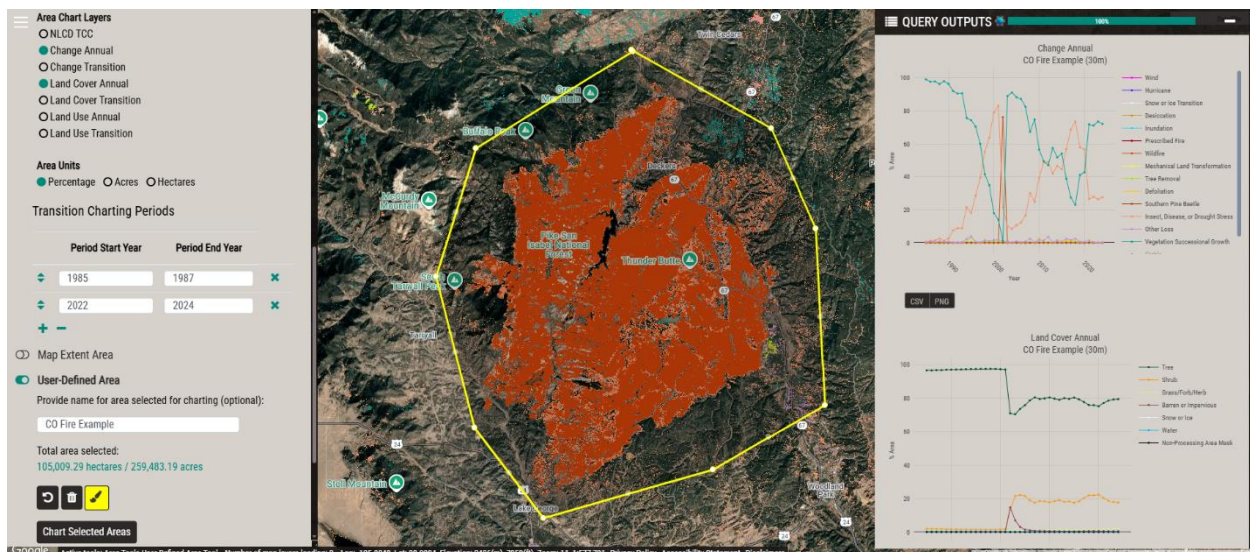
(a) Toggle on the User-Defined Area tool and type a name into the **Name your charting area** box, if desired.

(b) If desired, choose a color for the charting area by clicking the **paintbrush tool**.



(c) **Outline** the area of interest by clicking to create vertices for a polygon. **Double-Click** to finish drawing the boundary. If another polygon is wanted, repeat this process. Then click **Chart Selected Areas**.

(d) Charts will populate for the selected area in the **QUERY OUTPUTS** pane.



(e) Charts can be downloaded as a CSV or PNG using the buttons at the bottom left of each chart.

(f) To delete a user defined area of interest and draw a new one, click the **trash can icon**.



- iv. **Upload an Area:** The user can input a zipped shapefile (with all the necessary files) or geoJSON file to summarize across. Large shapefiles (vectors > ~5000 vertices or file size > ~50 mb) will likely fail. If this occurs, try increasing the **Vertex Reduction Factor** (maintains every nth vertex. E.g. if set to 3, every third vertex will remain). If this still fails, please contact us to summarize across a large vector.

☒ Upload an Area

Choose a zipped shapefile, kml, kmz, or geoJSON file to summarize across. Then hit "Chart across chosen file" button below to produce chart.

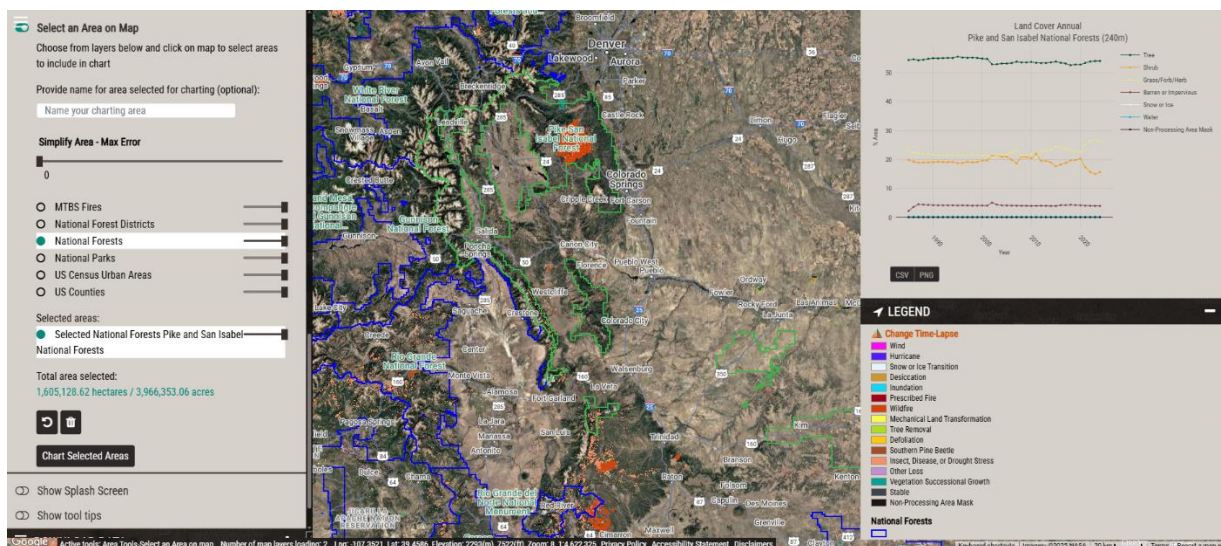
No file chosen

Vertex Reduction Factor

1

Uploaded areas:

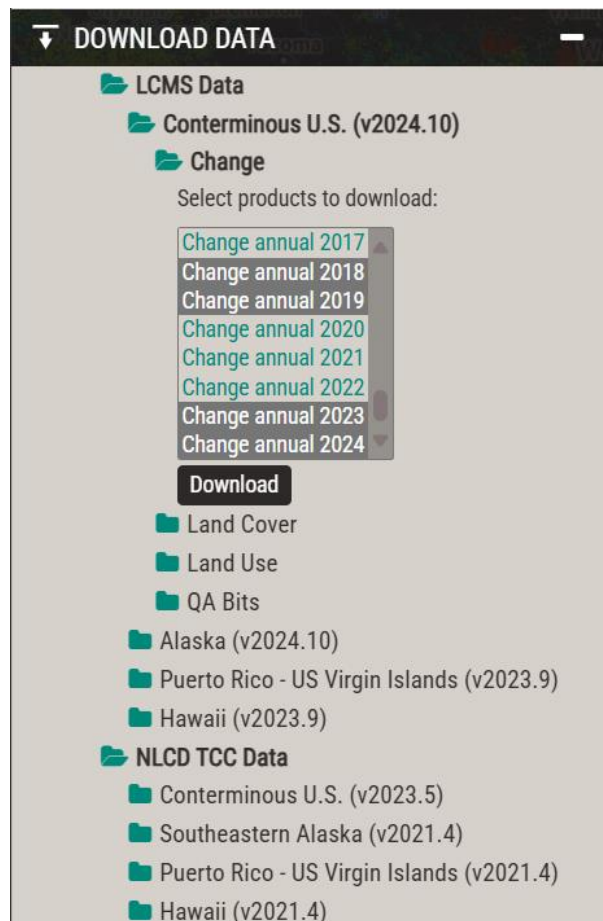
- (a) Click **Choose File** to insert a user shapefile (zipped in a .zip file) or geoJSON. Click **Chart across chosen file** button.
- v. **Select an Area:** The user can select from a list of pre-defined areas to generate the same summaries outlined above.
- (a) Use the radio buttons next to the list of layers to turn on a layer with pre-determined boundaries and click the polygons to include in the time series chart. The "Simplify Area - Max Error" slider can be used to simplify areas that are selected by allowing for a specified maximum error in meters. This allows larger and/or more complicated areas to be summarized. This simplification is performed as an area is selected. The selected areas added to the map reflect the simplification. You will notice the differences are usually quite minimal. If an area fails to run, clear the current selection, increase the value on the slider, and reselect the area(s). Click **Chart Selected Areas** to chart.



Part 3: Download Data & Support Tabs

A. Download Data tab

1. Click on the **Download Data** tab.
2. LCMS and TCC products can be downloaded for the conterminous United States, Alaska (Southeast Alaska only for TCC), Hawaii and Puerto Rico – US Virgin Islands. **WARNING!! – Downloads for the Conterminous United States are about 1GB per zip file. QA Bit zip files can be ~5gb.**
3. Select/highlight the products to download by dragging your cursor, holding down ctrl or shift, then click on download.
4. Disable the internet browser pop-up blocker before downloading products.
5. All downloaded LCMS products include the geospatial data in a geoTiff format (.tif), pyramid file (.tif.ovr), class names (.tif.aux.xml), html metadata (.tif.html), ESRI metadata (.tif.xml).



B. Support tab

1. The tutorial, methods documents, acknowledgements, and information for contacting the LCMS team is available here.



Note: LCMS and TCC data are not spatially and temporally equivalent in all cases. For example, TCC data are only available for Southeast Alaska, while LCMS is available for all of Alaska.

Congratulations! You have successfully completed this exercise. You now know how to navigate the LCMS Data Explorer and download data for use within your own project.